

Elaine Wu

Princeton University | Operations Research and Financial Engineering

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EDUCATION

Princeton University | B.S.E. Operations Research and Financial Engineering (ORFE) Expected Grad. May 2028

- **Minors:** Statistics & Machine Learning, Finance
- **Relevant Coursework:** Probability & Stochastic Processes, Optimization, Financial Risk Management, Data Intelligence, Data Structures & Algorithms, Energy & Commodities Markets, Mathematical Micro, Tokenized Finance
- **Activities:** The Daily Princetonian (Web Design and Development), Service Focus (Public Interest Technology)

Mountain Lakes High School | Unweighted GPA 4.0/4.0, Weighted GPA 4.879/4.0, SAT 1560 Sep 2020 - Jun 2024

- **Awards:** 2x AIME Qualifier (Top 2.5% with Distinction), National Merit Finalist, AP Scholar with Distinction
- **Activities:** Student Body VP; Editor-in-Chief for the school newspaper; 2x Captain, 4x Varsity Tennis Team

EXPERIENCE

Varsity Software | *Software Engineering Intern* Jan - Sep 2026

- Owned end-to-end product development of client web apps in JavaScript and Python, integrating Anthropic and OpenAI APIs into an internal tool to automate code and UI component generation
- Scaled [BizScoreLine](#), an AI-powered business assessment tool, from 2,000 to 10,000+ users through performance optimizations validated via A/B testing; authored API documentation (OAuth2, schemas, and error handling)

Susquehanna International Group | *Discovery Program (Quant Trading Track)* Aug 2026

- One of ~80 selected to apply statistics, game theory, and market-making concepts through trading simulations

Smile Train | *Data Analyst Intern* Jun - Aug 2025

- Built Qlik Sense dashboards from complex program data, enabling 15% faster funding reallocation
- Queried large SQL datasets to identify donor drop-off patterns, informing regional engagement strategy
- Defined governance policies that ensured data continuity during a QlikView-to-Qlik Sense migration

Jane Street | *WiSE Program* Aug 2024

- Selected among 70 nationwide to simulate market-making strategies, analyzing bid-ask spreads and order flow

Honors Summer Math Camp at Mathworks | *Researcher, Counselor* Jun - Aug, 2021-2024

- Optimized 4 statistical distance metrics in R; best metric achieved 89% predictive accuracy
- Developed constrained spectral clustering (ML) algorithm in Python and MATLAB; reduced runtime by 22%
- Communicated complex technical concepts to non-technical audiences at annual symposiums

PROJECTS

Stress-Testing 101 Formulaic Alphas | *Alpha Research, Overfitting Analysis, pandas*

- Re-implemented 13 WorldQuant alphas on 505 S&P 500 stocks and stress-tested 52 data-mined constants; published values beat random draws only 49% out-of-sample, and only the simplest alpha held up (IC t-stat 3.2 IS, 1.8 OOS)

Blackjack Card Counting Simulator | *Kelly Criterion, Monte Carlo Simulation*

- Simulated 1M+ hands of Kelly-sized card counting; grew a \$10K bankroll to a median 75x vs. 100% bust rate for flat betting, +1.97% edge at true count +4

GTO Poker Range Calculator | *Game Theory, Nash Equilibrium*

- Built a GTO push/fold calculator visualizing Nash equilibrium ranges at short-stack depths and EV reasoning behind shove/fold decisions

SKILLS AND INTERESTS

Programming Languages: Python, R, Java, SQL

Visualization and Tools: Excel, Qlik Sense, Salesforce, Figma

Interests: Poker, Tennis, Visual Art